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Motivation

- Long CoT improves reasoning but autoregressive decoding makes latency grow with every token.
- Over 98% of analyzed long reasoning traces contain parallelizable branches, about seven per example.
- Standard AR models rarely expose these branches explicitly and hidden-state probes are near chance.

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Full stack

DATA + MODEL + ENGINE

- Curator converts sequential CoT into validated MapReduce trees; Multiverse-1K contains 1,000 examples.
- A control-tag interpreter in SGLang forks paths, shares prefix KV cache and schedules them as a batch.
- Radix cache merges KV indices without physical copying or padding, preserving full branch context.

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Model

MAP -> PROCESS -> REDUCE

- Map sequentially writes a goal and indexed subtask outlines.
- Process generates each independent path concurrently, including nested collective or selective branches.
- Reduce concatenates all completed branch states and synthesizes a conclusion before sequential generation resumes.

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Evidence

QUALITY WITH LOWER LATENCY

- SFT of Qwen2.5-32B takes 3 hours on 8 B200s; AIME24/25 pass@1 reaches 53.8%/45.8%.
- Results match autoregressive-32B across AIME, MATH500 and GPQA while generating about 1.15-1.18x in parallel.
- Most samples average 18.5% speedup; highly parallel cases reach about 2.1x, stable up to batch 128.

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Attention

ISOLATED BRANCHES

- Multiverse Attention changes masks and position indices while remaining close to causal attention.
- Paths begin at the same position and cannot attend to one another during parallel execution.
- The Reduce position follows the longest branch, preventing invalid negative relative distances.

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Meaning & limits

TAKEAWAY

- Parallelism follows logical independence rather than brute-force simultaneous token prediction.
- The model can switch between sequential and parallel modes and scales more reasoning within fixed latency.
- Only SFT and 1K curated examples are studied; broader tasks and RL methods to encourage more parallelism remain open.

MULTIVERSE

Natively Parallel Generative Modeling

MapReduce Reasoning beyond Autoregression | 2025

Internalize a MapReduce program inside the language model: decompose tasks, generate independent reasoning branches in parallel, then merge their full KV states without information loss.

parallel decoding

MapReduce

reasoning